LAB EXPERIMENT

COURSE CODE: ITA0318

COURSE NAME: MOBILE COMPUTING

REG NO: 192324235 **NAME:** JOTHIKA M

EXP -16

16. Develop a wireframe for a mobile application that addresses challenges related to ad hoc wireless networks using Figma.

Aim: -

Create a Wireframe for a mobile application that addresses challenges related to ad hoc wireless networks using Figma.

Procedure: -

1. openFigma

2. Create a new file

3. Select the Frames

4. Design Visual Elements

5. Make it Interactive

6. Add icons on the Frame

7. Incorporate Multimedia

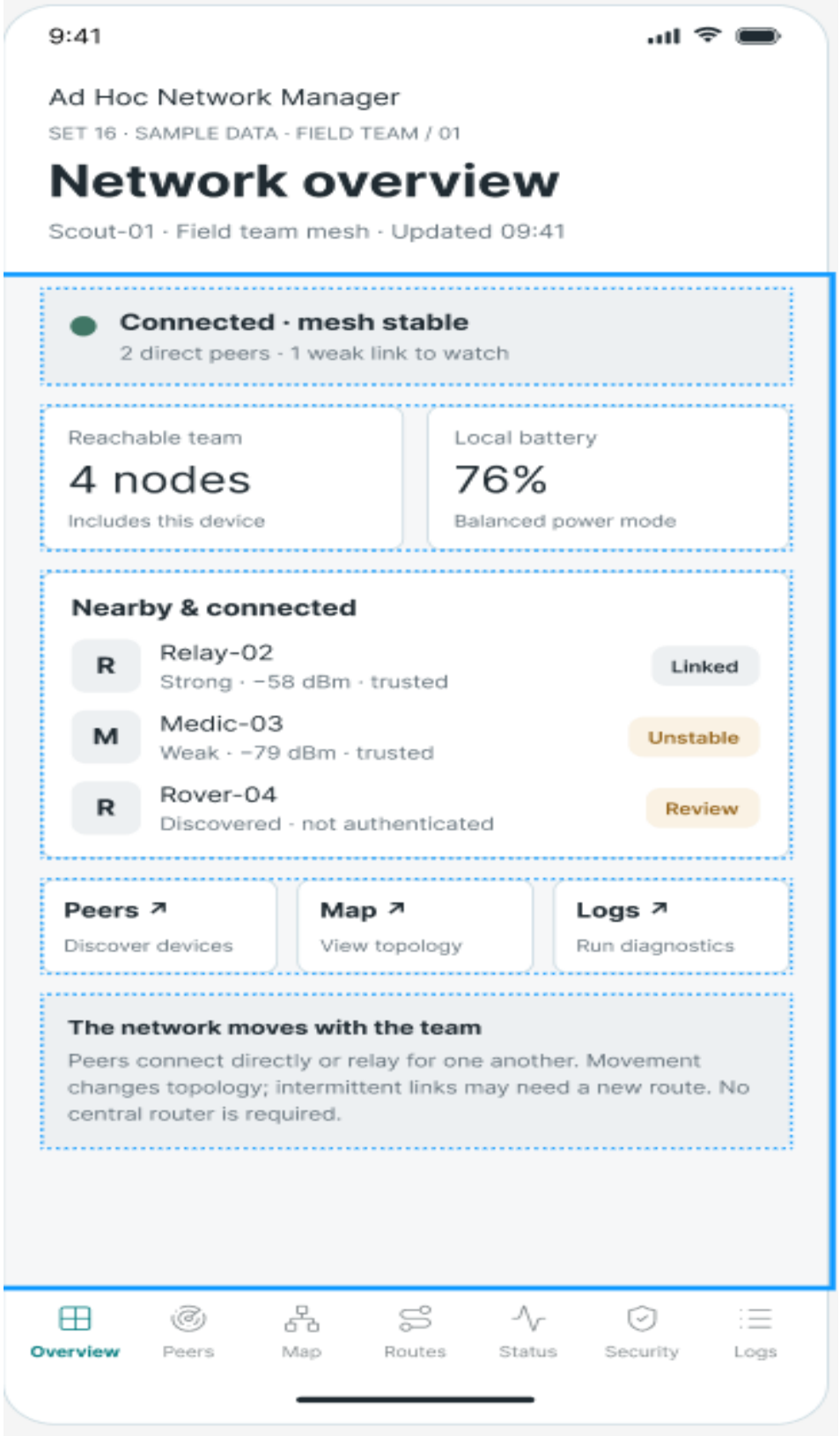
8. Storyboard Animation

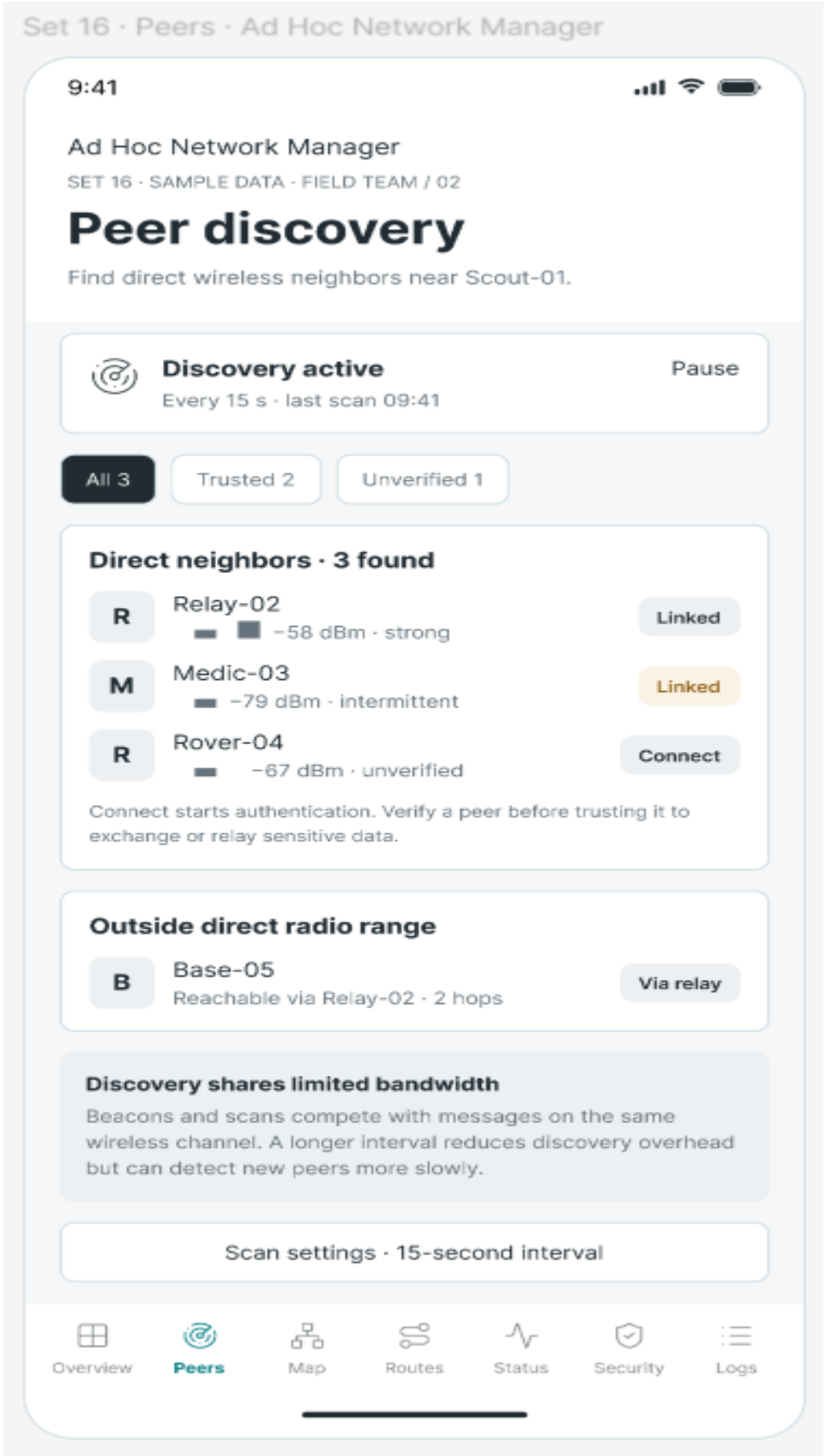
9. Review and edit the Prototype

10. Save and Share

Design:

Set 16 · Overview · Ad Hoc Network Manager




Set 16 · Peers · Ad Hoc Network Manager
9:41
Ad Hoc Network Manager
SET 16 · SAMPLE DATA · FIELD TEAM / 02
Peer discovery
Find direct wireless neighbors near Scout-01.
Discovery active
Every 15 s · last scan 09:41
Pause
All 3
Trusted 2
Unverified 1
Direct neighbors · 3 found
R
Relay-02
−58 dBm · strong
Linked
M
Medic-03
−79 dBm · intermittent
Linked
R
Rover-04
−67 dBm · unverified
Connect
Connect starts authentication. Verify a peer before trusting it to exchange or relay sensitive data.
Outside direct radio range
B
Base-05
Reachable via Relay-02 · 2 hops
Via relay
Discovery shares limited bandwidth
Beacons and scans compete with messages on the same wireless channel. A longer interval reduces discovery overhead but can detect new peers more slowly.
Scan settings · 15-second interval
Overview
Peers
Map
Routes
Status
Security
Logs

Set 16 · Map · Ad Hoc Network Manager

9:41

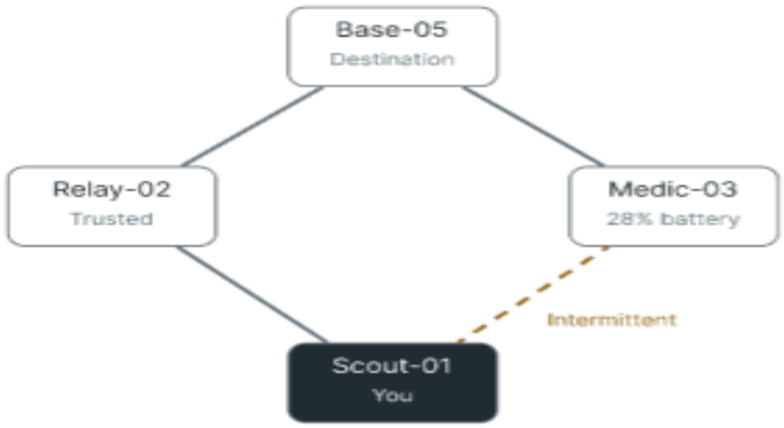
Ad Hoc Network Manager

SET 16 · SAMPLE DATA · FIELD TEAM / 03

Network topology

A live-style snapshot of peer-to-peer wireless links.

Field team · 4 reachable nodes



— Stable wireless link - - Intermittent link

Scout-01 is this device. Lines are wireless peer links, not infrastructure connections.

Not part of the trusted mesh

R

Rover-04

Discovered nearby · no mesh link

Unverified

Dynamic topology

Medic-03 is moving out of Scout-01's range. Its dashed link may disappear, while other peer links can appear as the team moves.

Recenter map

Refresh snapshot

Overview Peers Map Routes Status Security Logs

Set 16 · Routes · Ad Hoc Network Manager

9:41

Ad Hoc Network Manager

SET 16 : SAMPLE DATA : FIELD TEAM / 04

Message routing

Source: Scout-01 → Destination: Base-05

Message M-104 · delivered via 2 hops



Teal arrows = active message path · peers relay wirelessly

Hop sequence

Scout-01 → Relay-02 → Base-05

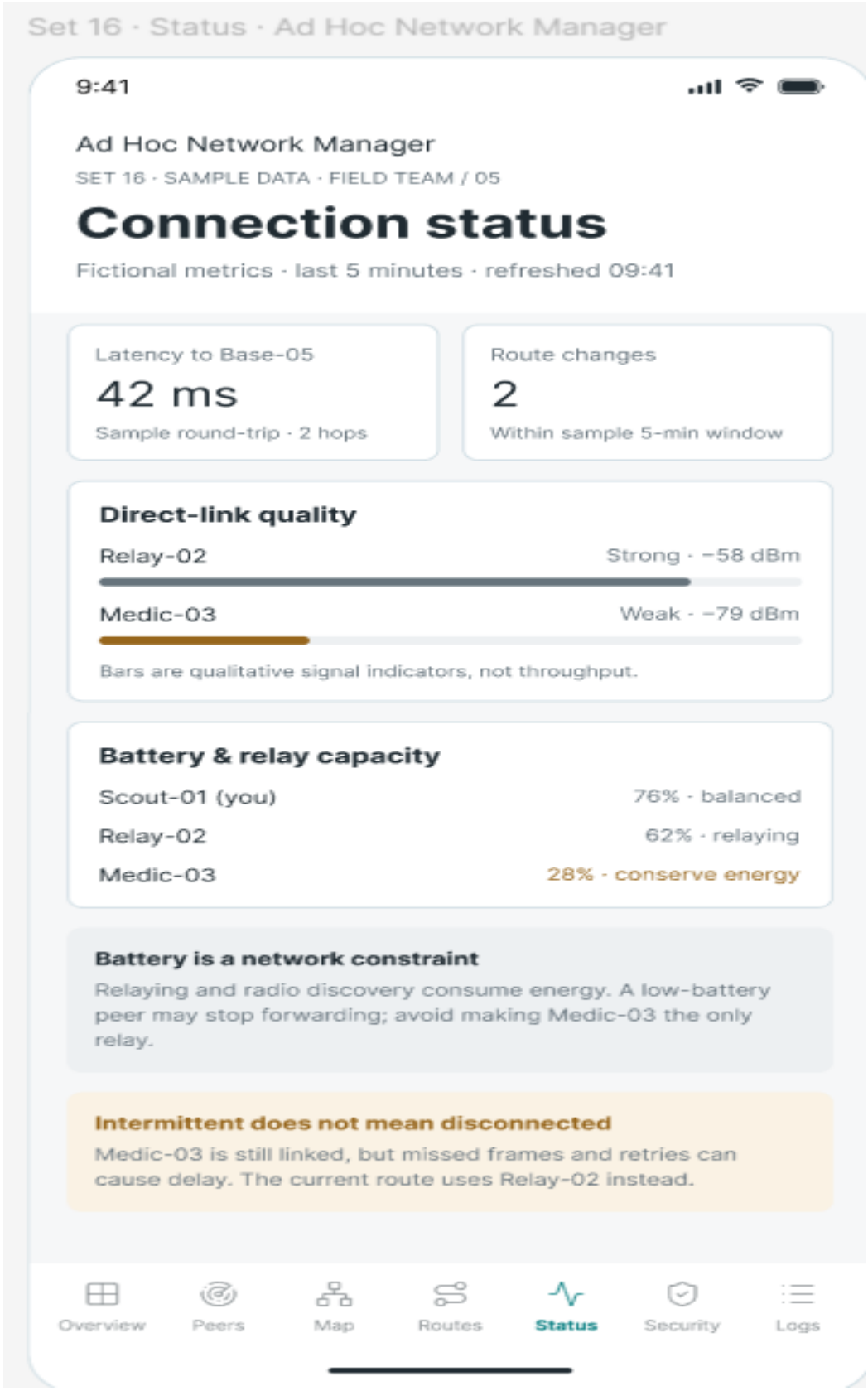
Alternate: Scout-01 → Medic-03 → Base-05. Not selected: Scout–Medic link is intermittent and Medic battery is low (28%).

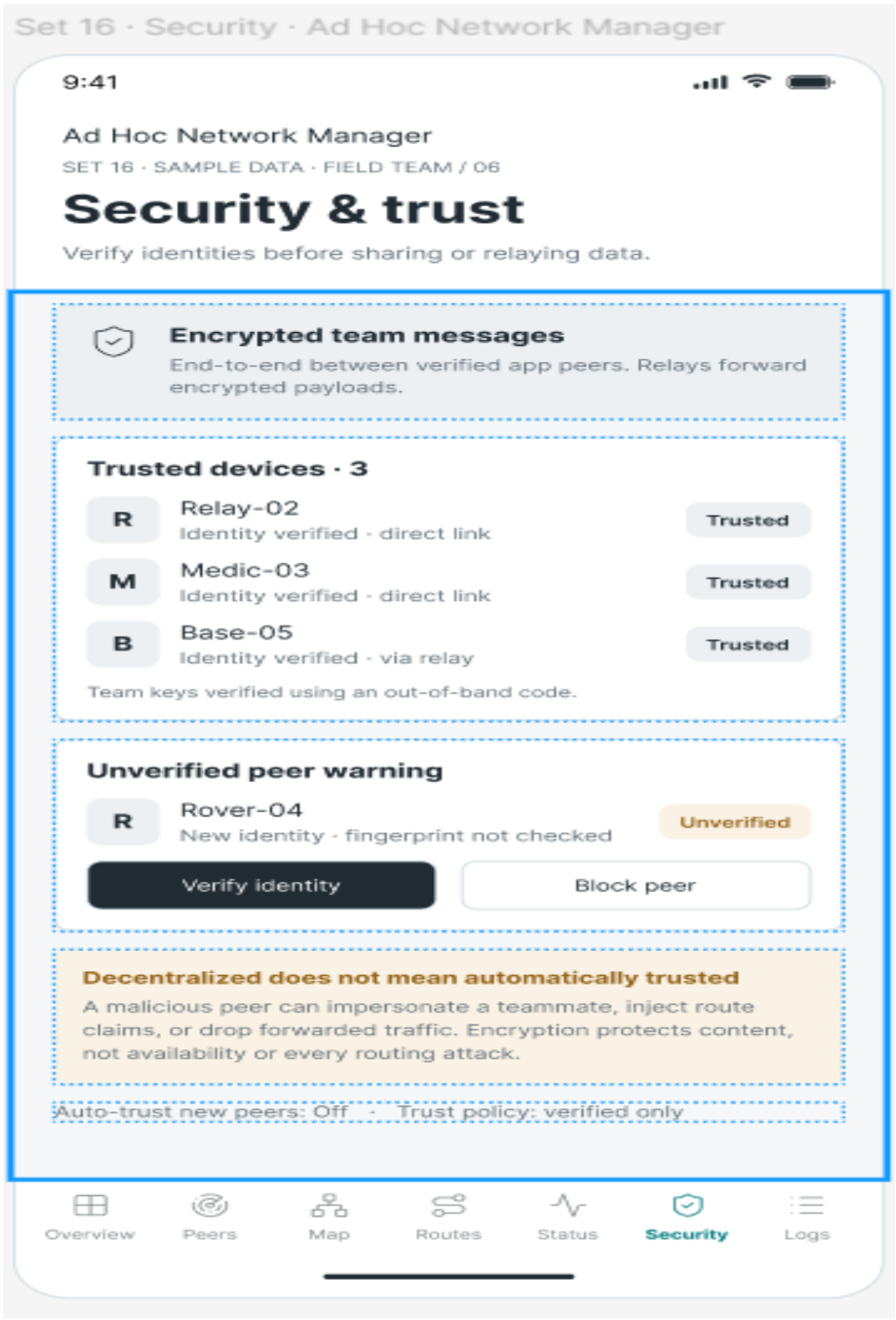
Route changes add overhead

Peer movement can break a hop. Route discovery and repair use channel bandwidth and energy, and may delay a message while a new path is found.

Route policy — Prefer stable links ›

Overview | Peers | Map | **Routes** | Status | Security | Logs


Set 16 · Status · Ad Hoc Network Manager
9:41
Ad Hoc Network Manager
SET 16 · SAMPLE DATA · FIELD TEAM / 05
Connection status
Fictional metrics · last 5 minutes · refreshed 09:41
Latency to Base-05
42 ms
Sample round-trip · 2 hops
Route changes
2
Within sample 5-min window
Direct-link quality
Relay-02
Strong · −58 dBm
Medic-03
Weak · −79 dBm
Bars are qualitative signal indicators, not throughput.
Battery & relay capacity
Scout-01 (you)
76% · balanced
Relay-02
62% · relaying
Medic-03
28% · conserve energy
Battery is a network constraint
Relaying and radio discovery consume energy. A low-battery peer may stop forwarding; avoid making Medic-03 the only relay.
Intermittent does not mean disconnected
Medic-03 is still linked, but missed frames and retries can cause delay. The current route uses Relay-02 instead.
Overview
Peers
Map
Routes
Status
Security
Logs


Set 16 · Security · Ad Hoc Network Manager
9:41
Ad Hoc Network Manager
SET 16 · SAMPLE DATA · FIELD TEAM / 06
Security & trust
Verify identities before sharing or relaying data.
Encrypted team messages
End-to-end between verified app peers. Relays forward encrypted payloads.
Trusted devices · 3
R
Relay-02
Identity verified · direct link
Trusted
M
Medic-03
Identity verified · direct link
Trusted
B
Base-05
Identity verified · via relay
Trusted
Team keys verified using an out-of-band code.
Unverified peer warning
R
Rover-04
New identity · fingerprint not checked
Unverified
Verify identity
Block peer
Decentralized does not mean automatically trusted
A malicious peer can impersonate a teammate, inject route claims, or drop forwarded traffic. Encryption protects content, not availability or every routing attack.
Auto-trust new peers: Off · Trust policy: verified only
Overview
Peers
Map
Routes
Status
Security
Logs



Result: -

Hence a wireframe for a mobile application that addresses challenges related to ad hoc wireless networks using Figma is created and executed successfully